

PYTHON FOR PLATFORM ENGINEERS · ISSUE 04 OF 04

Python reports for cost, inventory and incidents.

Automate the facts. Keep humans on the judgment.

04

AR

Akhilesh Ranjan Singh

Senior DevSecOps Engineer · github.com/iarsingh

Swipe →

One pattern for every report

- 1 Collect from APIs
- 2 Normalize
- 3 Enrich with owner tags
- 4 Summarize
- 5 Publish

Publish where people already look: chat, email or a markdown file in the repo.

Weekly cost movers

PYTHON

```
import pandas as pd

df = pd.read_csv("billing_export.csv")
df["owner"] = df["labels.owner"].fillna("UNTAGGED")

weekly = (df.groupby(["owner", "week"])["cost"]
          .sum().unstack())
last, prev = weekly.columns[-1], weekly.columns[-2]
weekly["delta_pct"] = (weekly[last] / weekly[prev] - 1) * 100

print(weekly.sort_values("delta_pct", ascending=False)
      .head(5))
```



Reports worth automating

REPORT	KEY CONTENT
Cloud cost	By owner, untagged spend, top movers
Inventory	Clusters, versions, end-of-life dates
Exposure	Public IPs, open buckets
Incident timeline	Alerts, deploys, changes in order



Incident summaries

01 Script assembles facts

Alerts, deploys, config changes, in time order.

02 Human writes the why

Root cause and decisions need judgment.

03 Links, not copies

Point to dashboards and logs.



Report habits

DO

- ✓ Flag untagged resources loudly
- ✓ Show week-over-week change
- ✓ Keep raw data for audit

AVOID

- ✗ Reports nobody acts on
- ✗ Manual copy-paste steps
- ✗ Generated root causes

KEY TAKEAWAY

Let scripts gather the facts so people can make the calls.

- Collect, normalize, enrich, summarize, publish
- Owner tags make cost actionable
- Humans write the why

YOUR TURN

Which weekly report would you automate first?

FOLLOW AKHILESH RANJAN SINGH

Series complete — save it for your next review

Save · Repost

